

Sample: Chapter 2, Sections 2.3–2.4 of the rebuilt report

Cross-references to sections not yet written show as ??; ignore them.

1.2 Grouping the jobs

Section ?? settled the loading problem. What remains is the order. Solving for the order directly is hard, so practice does something indirect. It first asks a simpler question: which jobs could share a magazine?

Definition 3 (Feasible group). A set of jobs $G \subseteq J$ is a *feasible group* if the tools its jobs need, taken together, fit in the magazine:

$$\left| \bigcup_{j \in G} T_j \right| \leq b.$$

A feasible group can be run start to finish without a single tool change. Load the tools its jobs need, run them in any order, and only then change anything. This makes the following question natural.

Definition 4 (Job Grouping Problem). The *Job Grouping Problem* (JGP) is the problem of partitioning J into as few feasible groups as possible. The smallest possible number of groups is written K^* .

The JGP ignores order completely. It asks only how the jobs may be divided. It is itself NP-hard, but it is much easier in practice than the SSP, and it can be solved exactly on the instance sizes considered here.

Two problems now sit next to each other. The JGP says which jobs may share a magazine. The SSP asks for an order. The standard heuristic joins them, in three steps.

1. **Group.** Solve the JGP. This returns K^* feasible groups $G_1, \dots, G_{K^*}$, and for each group a magazine *configuration* C_l of size b containing everything the group needs.
2. **Sequence.** Decide the order of the groups. The cost of moving from configuration C to configuration C' is the number of tools that must be inserted, $d(C, C') = |C' \setminus C|$. Finding the cheapest order of the K^* configurations is a travelling-salesman problem on K^* points, which is small enough to solve exactly.
3. **Evaluate.** Write out the jobs group by group, in the order chosen in step 2, to obtain a single job order. Apply KTNS to that order and report its cost.

This is the method studied throughout this report. It is fast, it is what practitioners use, and its output is always a valid schedule, so its cost is never below the optimum Z^* .

Example 2 (The heuristic on the 6-ring). Take the 6-ring of Example ??: six jobs, six tools, $b = 3$, and $T_j = \{j, j + 1\}$ with indices modulo 6.

Step 1 returns $K^* = 3$. One optimal grouping is

$$G_1 = \{1, 2\}, \quad G_2 = \{3, 4\}, \quad G_3 = \{5, 6\},$$

whose tool requirements are $\{1, 2, 3\}$, $\{3, 4, 5\}$ and $\{5, 6, 1\}$. Each has exactly three tools, so each group fixes its configuration with no freedom left: $C_1 = \{1, 2, 3\}$, $C_2 = \{3, 4, 5\}$, $C_3 = \{5, 6, 1\}$.

Step 2 compares the three configurations. Any two of them share exactly one tool, so moving between any two costs two insertions. Every order of the three groups therefore costs the same, and step 2 may return any of them. Take G_1, G_2, G_3 .

Step 3 writes out the jobs group by group, giving the order 1, 2, 3, 4, 5, 6, and applies KTNS to it. That is the order already traced in Example ??, and its cost is 6 insertions, or 3 once the first fill is not charged.

Two things in this example are worth pausing on, because both matter later.

First, the method does not fully determine its own answer. Step 1 may return several optimal groupings; step 2 may face ties, as it does here; and the order of the jobs *inside* a group is left open by all three steps. Different choices can give different costs. Throughout this report the heuristic is read in its most favourable form: its cost is the best value reachable over all these choices. This is the reading that makes the comparison with Z^* meaningful, because any weaker reading would measure the tie-breaking rule rather than the method.

Second, and less obviously, step 3 does something that step 2 did not anticipate. This is the subject of the next section.

1.3 Two ways to measure the heuristic

Steps 1 and 2 reason entirely in terms of configurations. A group is assigned one magazine, and that magazine is imagined to stay loaded while the whole group runs. Under that picture the schedule is a walk through K^* configurations, and its cost is the sum of the transition costs $d(C, C')$ along the walk.

Step 3 does not do this. It discards the configurations, keeps only the job order they produced, and hands that order to KTNS. KTNS is free to change the magazine between any two consecutive jobs, including two jobs of the same group. The group structure survives only as an ordering of the jobs. It does not constrain the magazine.

The two readings give different numbers, and both appear in the literature without being distinguished. This report separates them.

Definition 5 (The walk value and the heuristic value). Let $\mathcal{P}$ range over the partitions of J into K^* feasible groups, and for each such partition let the configurations be chosen and ordered as cheaply as possible. Write

$$H_{\text{walk}} = \min_{\mathcal{P}} \min_{\sigma} \sum_{l=1}^{K^*-1} d(C_{\sigma(l)}, C_{\sigma(l+1)})$$

for the cheapest such walk. Write H for the cost of the two-phase heuristic as defined in Section 1.2: the minimum, over the same partitions, over all orders of the groups and all orders of the jobs within each group, of the KTNS cost of the resulting job order.

Proposition 2. *For every instance, $H \leq H_{\text{walk}}$.*

Proof. Fix a partition and an order of its groups attaining H_{walk} . Writing the jobs out group by group in that order gives a job order. Holding each group's configuration fixed while its jobs run is one admissible loading for that order, and its cost is exactly the walk cost. By Proposition ??, KTNS returns the cheapest loading for that order, so its cost is at most the walk cost. Since H minimises over at least this one job order, $H \leq H_{\text{walk}}$. $\square$

The inequality can be strict, and it is strict on the running example.

Example 3 (The two values differ on the 6-ring). Continue Example 2. The walk pays three insertions to build $C_1 = \{1, 2, 3\}$, then two to reach $C_2 = \{3, 4, 5\}$, then two to reach $C_3 = \{5, 6, 1\}$. That is $H_{\text{walk}} = 7$, or 4 free-initial.

KTNS on the same job order costs 6, or 3 free-initial (Example ??). Since $Z^* = 3$, the heuristic is *optimal* on the 6-ring, while the walk value is one above the optimum.

Figure 1 shows where the two part company. The walk insists that the middle group runs under $C_2 = \{3, 4, 5\}$ and nothing else. Tool 1 is not in C_2 , so it is evicted at position 3 and must be reloaded at position 5 for job 6. KTNS is not bound to C_2 . It holds $\{1, 3, 4\}$ at position 3 and $\{1, 4, 5\}$ at position 4 — two different magazines inside one group, each serving its own job — and so keeps tool 1 resident throughout. The one re-insertion is avoided, and the schedule is optimal.

Remark 1 (Why the distinction is kept throughout). It would be simpler to work with H_{walk} alone. It has a clean description — a shortest walk through K^* configurations — and the structural results of Chapter ?? are naturally proved about it.

That simplification is not available, for two reasons. The first is that H_{walk} is not what the method returns. A practitioner running the three steps of Section 1.2 obtains H , and any statement about how far the method can be from optimal must be a statement about H . The second is that the two can differ on exactly the instances that matter: on the 6-ring, the family this report returns to most often, H_{walk} exceeds the optimum while H attains it.

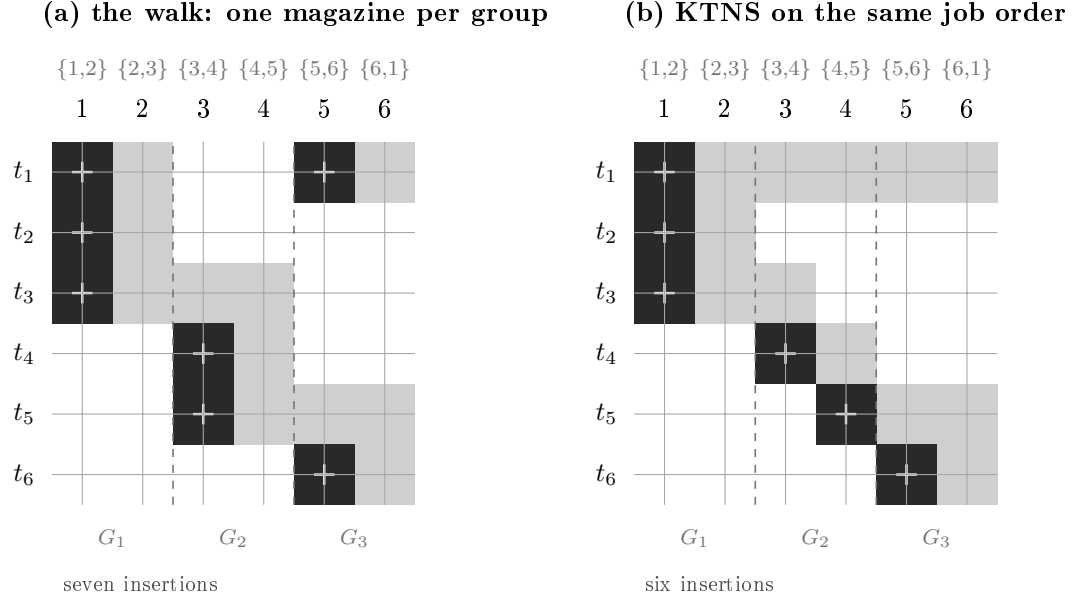


Figure 1: The same grouping of the 6-ring, measured two ways. Columns are the six processing positions, with the tools each job needs above them; rows are the six tools. A shaded cell means the tool sits in the magazine at that position; a “+” marks an *insertion*. The dashed lines separate the three groups. In (a) each group runs under one fixed magazine, so tool 1 (top row) is evicted at position 3 and reloaded at position 5 for job 6: seven insertions. In (b) KTNS may change the magazine inside a group, so tool 1 is loaded once and kept across all six positions, and the count is six. That single re-insertion is the whole difference between H_{walk} and H .

Every result in Chapter ?? is therefore stated with its value named. Proposition 2 carries upper bounds from one to the other: anything proved to bound H_{walk} from above also bounds H . Nothing transfers in the other direction, and in particular an instance on which H_{walk} exceeds Z^* is not, by itself, an instance on which the heuristic is suboptimal.